

AGRIAI 2.0 | FORENSIC SOFTWARE EXAMINATION

Verification Evidence and Reproduction

Commands, results, route observations, dependency evidence and source fingerprints

Volume 08

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Repository examined: ThaddeusHackz/agriai-v2

Evidence commit: 991457272b7fec548e02c645c5593b930af5ea29

Examination date: 18 August 2026

Classification: Internal technical record

Technical basis

This document is grounded in direct source review, route enumeration, dependency audit, secret pattern scanning, strict TypeScript checking, ESLint, production compilation, PostgreSQL integration tests and local HTTP behavior tests. Live third party AI accounts were not available in the examination environment, so provider success is assessed from code paths and contracts rather than a claim of live production availability.

1. Reproducible examination workflow

```
npm ci
npm run typecheck
npm run lint
npm run test:pg
npm audit --json
npm run build
npm run start -- --hostname 0.0.0.0 --port 3000
curl local route matrix
tracked-file secret pattern and entropy scan
source SHA-256 calculation
```

2. Build and code quality results

Command	Observed result	Forensic interpretation
npm ci	539 packages added, 540 audited by install	Lockfile resolved and install completed
npm run typecheck	Exit 0	Strict TypeScript check passed
npm run lint	Exit 0, 22 warnings, 0 errors	Build-quality warnings remain but lint does not fail
npm run test:pg	14 pass, 0 fail	Document mirror and hydration test suite passed
npm run build	Exit 0, 35 route/page entries	Production output compiled
Build diagnostics	4 Edge instrumentation warnings	fs, path, crypto and cwd are included in Edge analysis

3. Lint warning classes

Class	Count or examples	Impact
React set state in effect	Chat, CountUp, MarketPrices, Weather, admin tabs, SiteProvider	Possible cascading render and maintainability cost
Unused imports or variables	DiseaseDetector Upload, Nav X, Newsletter settings and others	Low direct risk, code cleanliness
Unoptimized img	DiseaseDetector	Performance and image optimization warning
Font loading	app/layout.tsx	Next lint convention warning

4. Production route manifest

The production build marked /, /admin and /admin/login as static pages. It marked all 27 /api route files as dynamic. The build also generated icon.png, robots.txt and sitemap.xml. This confirms that backend code is compiled as route handlers and not shipped as a separate service.

5. Local HTTP smoke evidence

Request	Status	Observed response
GET /	200	HTML shell, 96,191 bytes
GET /api/config	200	Public settings and provider flags
GET /api/health	200	All providers unconfigured in test environment
GET /api/weather?city=accra	200	Seed fallback after provider network failure
GET /api/prices	200	Twelve seeded rows
GET /api/images	200	Empty demo image result without key
POST /api/chat malformed	400	Invalid JSON body
POST /api/chat empty	400	Message is required
POST /api/chat valid	200	SSE done with local knowledge
GET /api/history known id	200	Full chat returned without auth
GET /api/admin/me no cookie	401	Unauthorized
GET /api/admin/settings no cookie	401	Unauthorized
POST /api/generate no key	503	Configuration error
POST /api/vision invalid	400	Valid image required
POST /api/transcribe no key	501	Gemini missing
POST /api/tts no key	501	ElevenLabs missing
POST /api/admin/login fallback	200	Known fallback credential issued cookie
GET /api/admin/me cookie	200	Admin identity returned
GET /api/admin/analytics cookie	200	Authenticated metrics returned
POST /api/admin/logout	200	Session removed

6. Response header evidence

```
HTTP/1.1 200 OK
X-Content-Type-Options: nosniff
Referrer-Policy: strict-origin-when-cross-origin
X-Powered-By: Next.js
Cache-Control: s-maxage=31536000
Content-Type: text/html; charset=utf-8
```

The header test confirms the two configured hardening headers and the framework fingerprint. It also confirms that CSP, frame restrictions and Permissions-Policy were absent in the local production response.

7. Dependency audit snapshot

Package	Severity	Relationship	Observed remediation
next 16.2.10	High	Direct	Audit recommends next 16.3.1
postcss <= 8.5.22	High	Transitive under next	Resolved through Next upgrade
sharp < 0.35.0	High	Transitive under next	Resolved through Next upgrade

The audit listed framework advisories involving proxy bypass, Server Action denial of service, SSRF conditions, cache confusion, image optimization and internal endpoint disclosure. Applicability varies because this source has no custom rewrite, no proxy file and no obvious Server Action declaration. A patched direct dependency remains the correct control.

8. Secret scan result

* No Google API key pattern, GitHub token, AWS access key, OpenAI key or PEM private key was found in tracked text.

- * Two database URLs matched credential syntax. One is the obvious .env.example placeholder. One is the local pg-mem test fixture. Neither is a live production secret.
- * High-entropy candidates were generated forensic PDF filenames, not credentials.
- * The scan cannot detect secrets that are encrypted, fragmented, stored only in Git history beyond the shallow checkout or held outside tracked files.

9. Key source fingerprints

File	SHA-256 at examined source state
package.json	575e919cc40665fed9db5a21a9e1b793899ac6bb4865937fc84b0b2044e50dae
package-lock.json	2f989b311325701f981fe1f62733ad4209d241298cd72082c049acb5d33d9797
next.config.ts	b05fc149510b1291c294d033529f0f41880783fa8ca1aadd293ffbe3dc0c0af7
render.yaml	64925e70485846aaf8c5030fd25359ef4e99e8302a72697f47eb9d43d6264d92
Dockerfile	003c4dcd38a97538cd24e1ffdfcde72c437d40d4ea726c6ac689a5a4481c33cb
app/api/chat/route.ts	6c5e7ee0d4dfba6e3ad913a5413c593de644373a9e791ada23a1a9b9cc8b79af
app/api/vision/route.ts	529a93154c356fd424ba2646a6631719cab70e4ccc825b5e35d7e3e695e7ec9c
app/api/generate/route.ts	b43bc2af3e142922d426bdaa87869bedabcbdd25f5993f3a0b18481a17b8a45e
app/api/transcribe/route.ts	61dc92aa631801efa3b282c8732e4a7baefbe6bb3f3c8398b2589745f5dc165f
app/api/tts/route.ts	17525c94c5735be8cae3217e319f8e3eb8b2789229468ad05c5cda1bd36cd457
app/api/history/route.ts	0c175f9699cec07dbba2d4bdfdd4fd4e6cac576e89bc5cd214b73a1c007e534a
app/api/health/route.ts	5ee9c3e7ecd6efdd1278ee6d71714b93821fdbd8cdd25019377ba161a2209b94
app/api/admin/login/route.ts	08d8d595d67cd4e7d974dc402b75dfd72bb5df149f78bc5436c8e93d508e224d
app/api/admin/settings/route.ts	937b2e86db7b74047a7ef4b946a2cbab9fe616af3556de61386a83b45b9929e6
lib/ai.ts	6acd98a42af534c9e047c64eae1d59af5bc20f7630e392142b2871e98981becb
lib/cloudflare.ts	a894b645b9acea3f272c0a58b9bf10c1c6bc4969c65df303b413d95376268521
lib/search.ts	8eddef4f6d6fbd62293391445ea1072b14d340b6fedeb86925b8d0abe3345b8a
lib/env.ts	c1c7263c1118fdad6f84c7a41dfacc426cded78ba55ae8fa68c33cd8adedd901
lib/db.ts	d462b1c53020cec809a29db5fbc417ba1ba24d0ae1169c46f8c10b44012ad548
lib/pg-store.ts	4fc90146fbdeef98b5bcfa30f3235578a828a6fe864c4f4b3045956c4c18f7e8
lib/auth.ts	ae7094e658273049e91a65c7a02a67d543855e06da086bfa2d053d6f7f70dfbe
lib/types.ts	4cc31b557f24a2b8f19ee47b27957c9b0ba73269817fb6809515b765a7503d79
components/Chat.tsx	5269862c8e8b7b63a2be79ada65548683e6727fcbabae8147e7beaf4f0694ba4

10. PostgreSQL test coverage

- * DATABASE_URL configuration detection.
- * Table initialization.
- * Version 4 seed and Studio section.
- * Sole-founder footer and prompt attribution.
- * Seed admin identity.
- * Forced JSONB upsert and load.
- * Seed version preservation.
- * Debounced save.
- * Health report.

- * Hydration that recreates a missing local file.
- * Database reset and reseed.

11. Missing test evidence

- * No provider-key integration test was run or found in the repository.
- * No test proves multilingual output quality or disease classification accuracy.
- * No load, race, memory or multi-instance consistency test exists.
- * No browser end-to-end test covers chat streaming, microphone, Studio or admin tabs.
- * No automated security test covers default credentials, IDOR, CSRF, RBAC or rate limits.

12. Reproduction discipline

Repeatability

Run all commands against the exact commit and preserve npm audit JSON with the examination date because advisory data changes over time. Provider tests should use dedicated low-quota test projects, never production keys. Clean the ignored data/db.json fixture after behavior tests.

Authorship and control statement

This volume was prepared, authored and issued solely under the name of Thaddeus Nii Teiko Tagoe. The command results and source fingerprints in this volume establish a repeatable evidence trail for the dossier. The repository records identify Thaddeus Tagoe as the founder, sole designer, developer and maintainer of AgriAI.